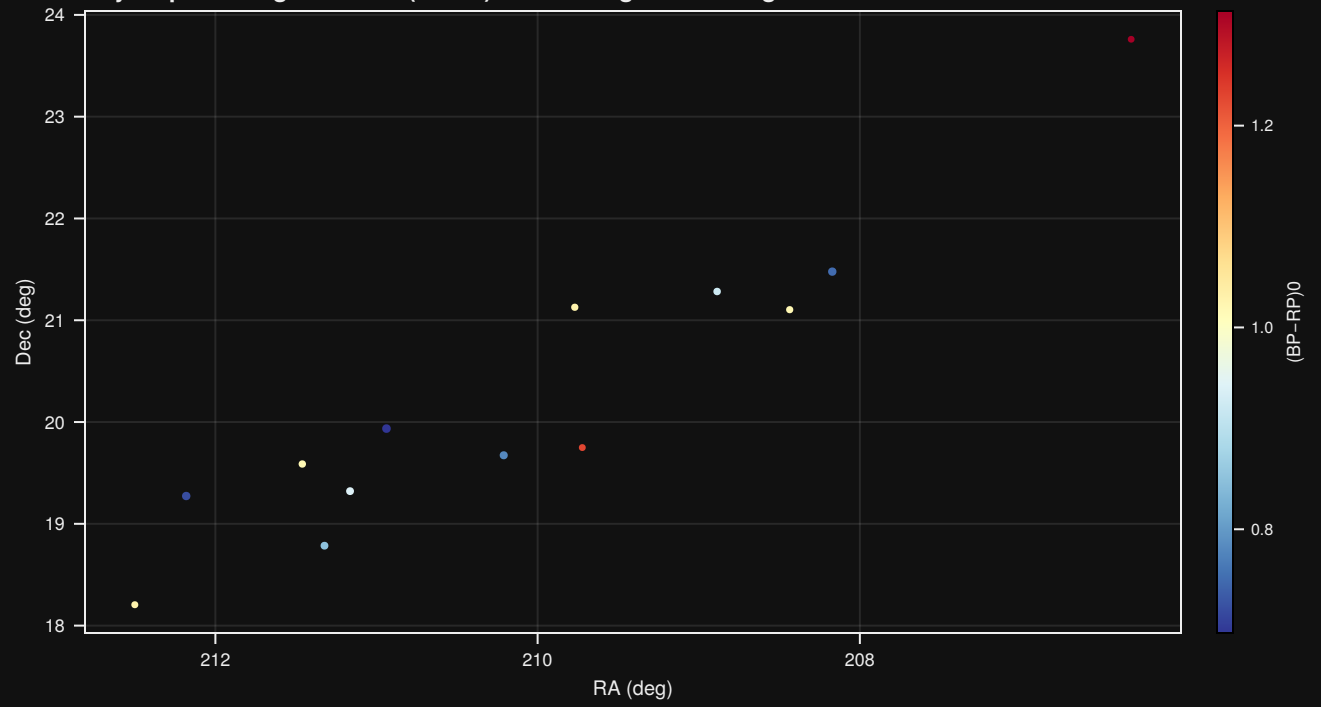
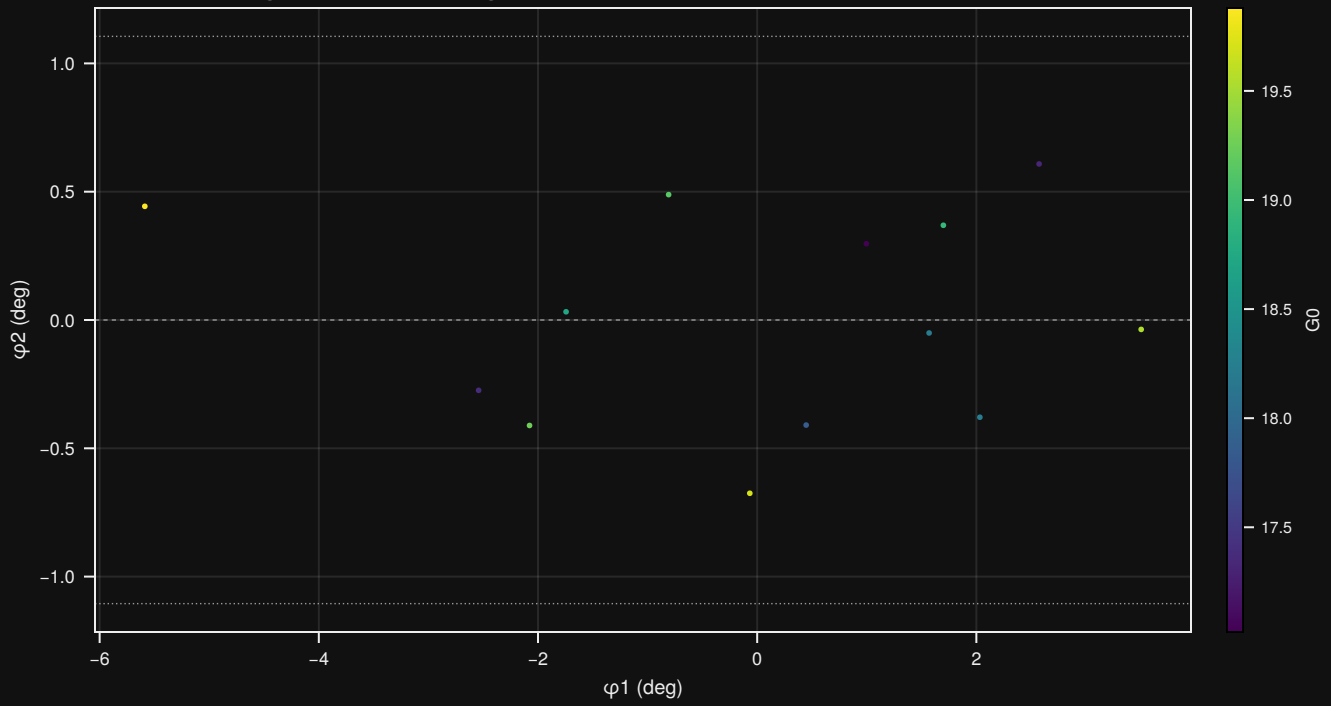
Ophiuchus

MEMBERSHIP	
Stars in catalog	13
with Gaia GSP-Phot	7 (54%)
with radial velocity	0
Stream id (sID)	56
GEOMETRY · great-circle fit	
Centroid RA, Dec	210.1, 19.7 deg
Centroid l, b	13.3, 72.4 deg
Pole RA, Dec	326.5, 50.3 deg
Length (φ1, 2–98%)	9.1 deg
Width σ(φ2), robust	0.55 deg
KINEMATICS	
pmRA* med ± σ	-23.04 ± 0.50 mas/yr
pmDE med ± σ	0.18 ± 0.31 mas/yr
μφ1* med ± σ	-19.14 ± 0.41 mas/yr
μφ2 med ± σ	-12.97 ± 0.31 mas/yr
Parallax (cat.) med	0.269 ± 0.265 mas
Parallax (wtd, ZP-corr.)	0.314 ± 0.047 mas
Distance	d = 3.2 ± 0.5 kpc
Physical length	0.50 kpc
RV median ± σ(MAD)	—
RV gradient dV/dφ1	—
PHOTOMETRY · POPULATIONS	
G0 range (median)	17.0–19.9 (18.7)
(BP−RP)0 median	0.94
GSP-Phot [M/H] (n=7)	-1.25 ± 0.42
GSP-Phot Teff median	5285 K
RUWE > 1.4	0.0 %
PM clusters (largest)	none
IBATA+2024 · TABLE 3	
n catalog | paper	13 | 391
n_RV catalog | paper	0 | 72
<[Fe/H]> paper	-1.98
Stellar mass (paper)	4800 Msun
Pole offset vs paper	46.6 deg
LABEL CHECK	n differs from paper!

Sky map · catalog members (n = 13) · size ∝ brightness · ring = has RV



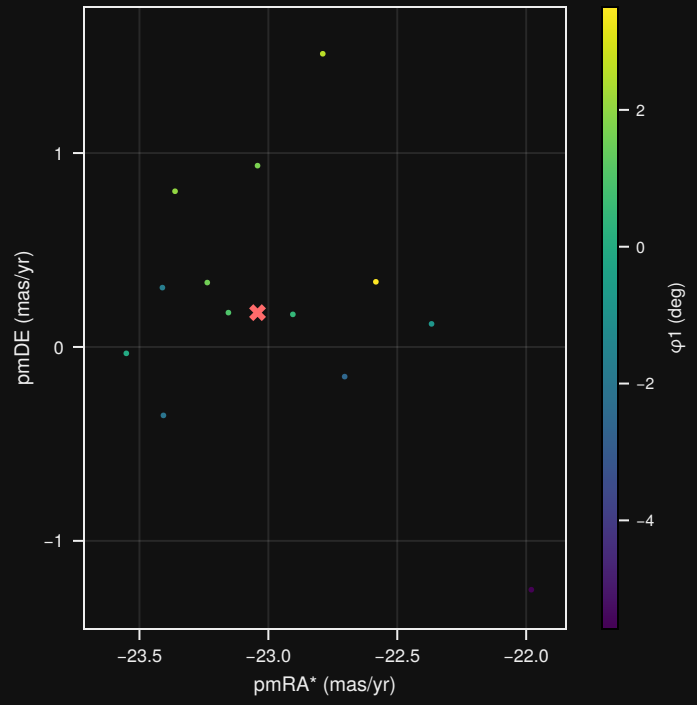
Stream frame (great-circle fit) · length 9.1° · width σ 0.55°



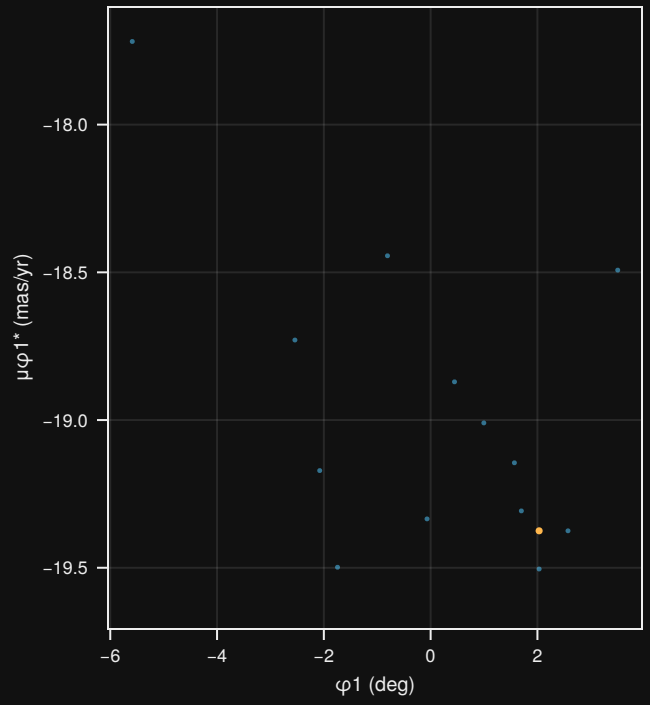
Ophiuchus

Kinematics · proper motion · parallax · radial velocity

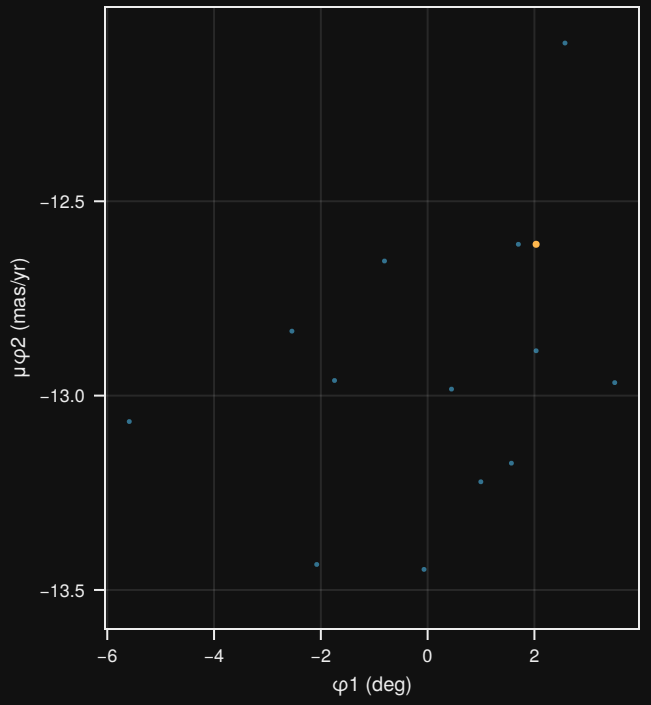
Proper motion (n = 13)



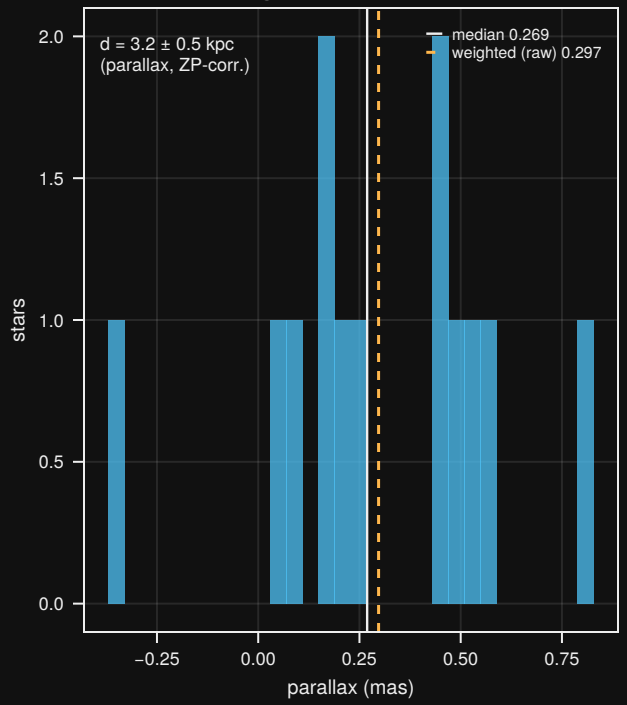
μφ1* vs φ1 · σ = 0.41



μφ2 vs φ1 · σ = 0.31



Parallax (catalog, n = 13)



Radial velocity vs φ1 (n = 0)

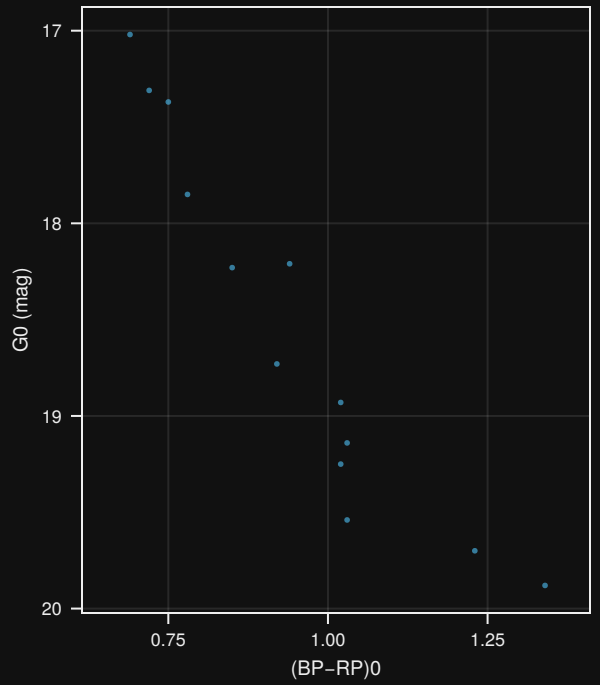
no radial velocities

V_hel distribution

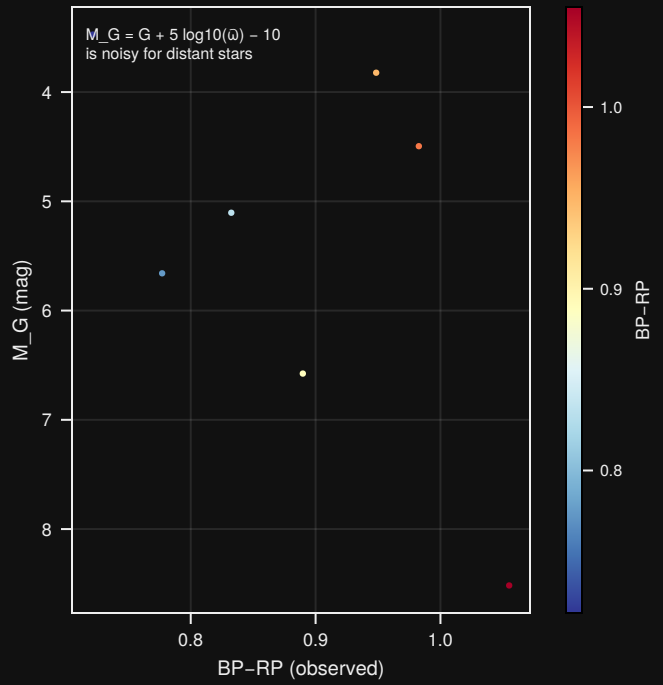
n_RV = 0 (< 3)

Ophiuchus

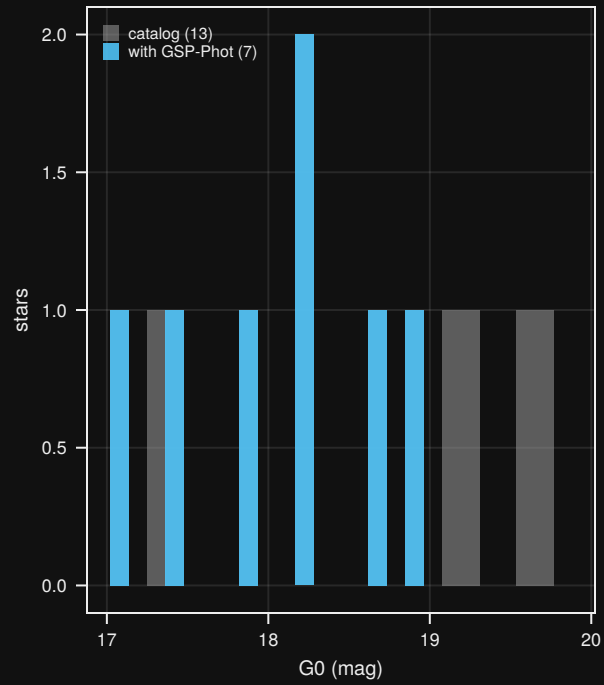
CMD · dereddened (n = 13)



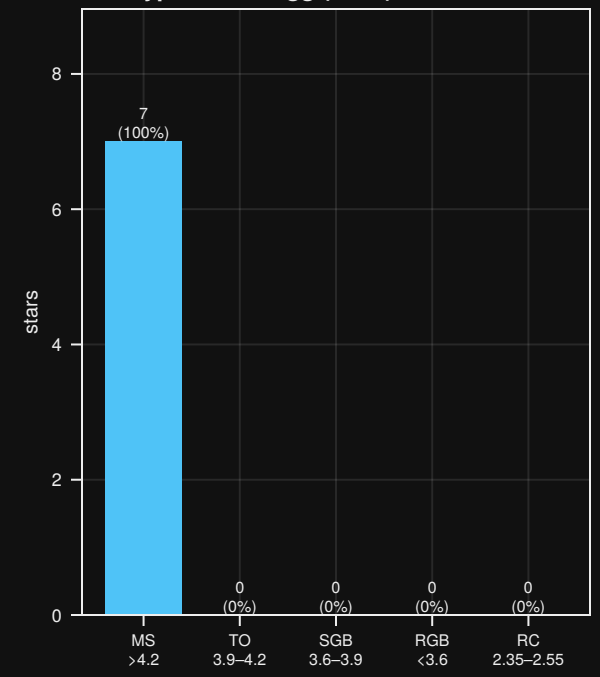
HR diagram · ϖ>0 (n = 7/7)



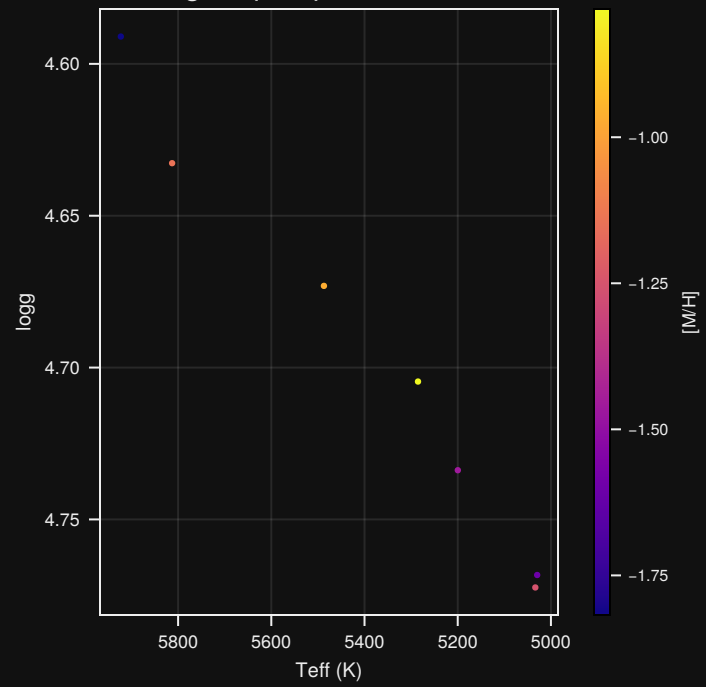
G0 & GSP-Phot coverage



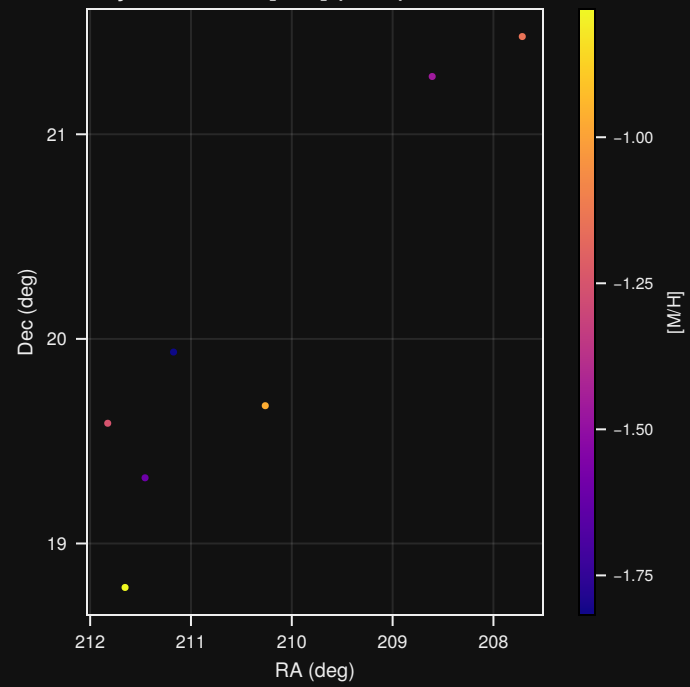
Star types from logg (n = 7)



Kiel diagram (n = 7)



Sky · GSP-Phot [M/H] (n = 7)

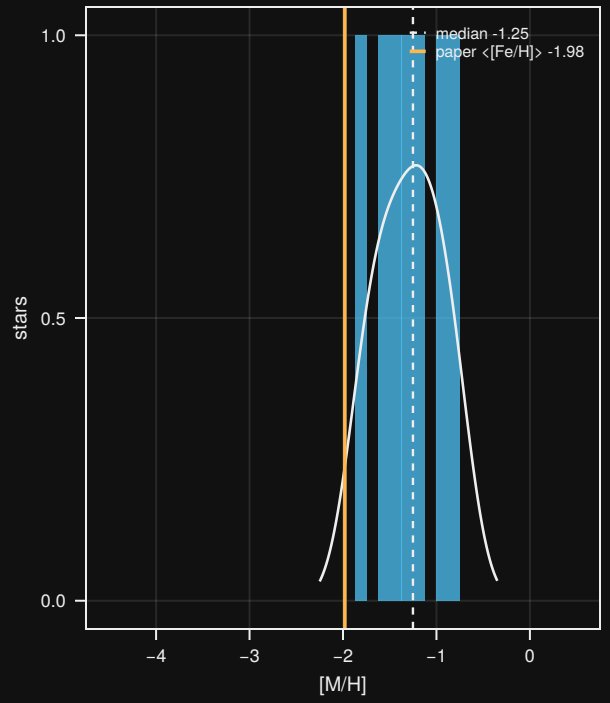


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

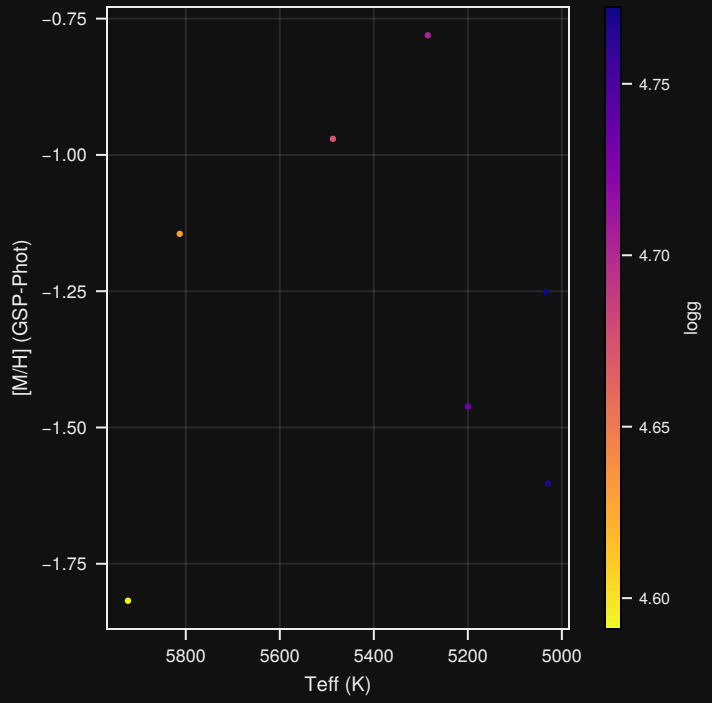
Ophiuchus

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 7)



Teff vs [M/H] (n = 7)



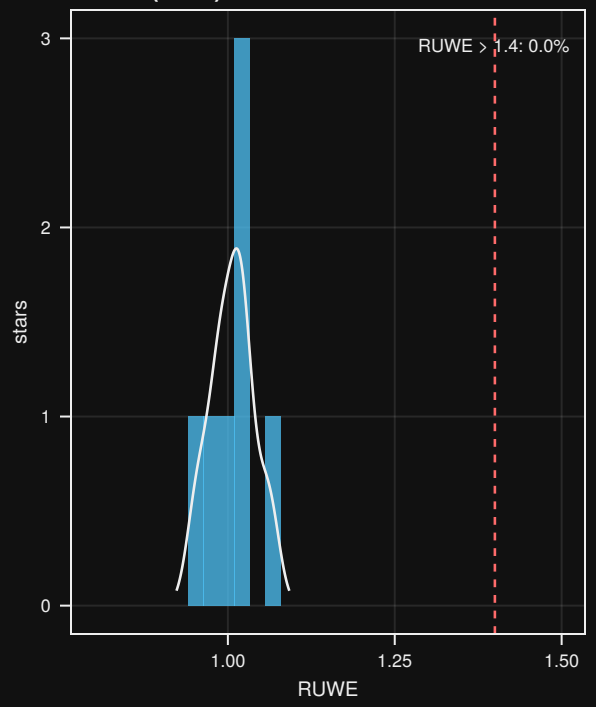
FLAME age (n = 0)

FLAME ages for only 0 star(s)
(FLAME needs good parallaxes)

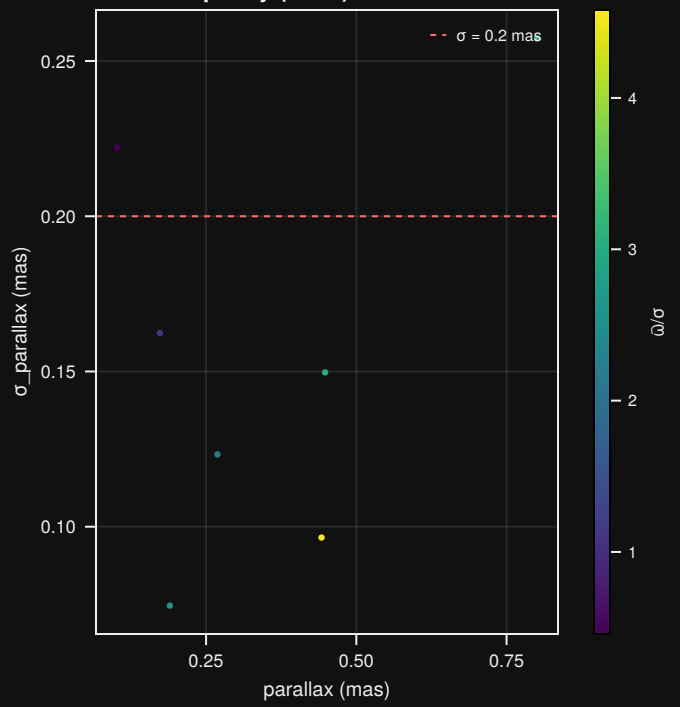
Age–metallicity (n = 0)

n < 3

RUWE (n = 7)

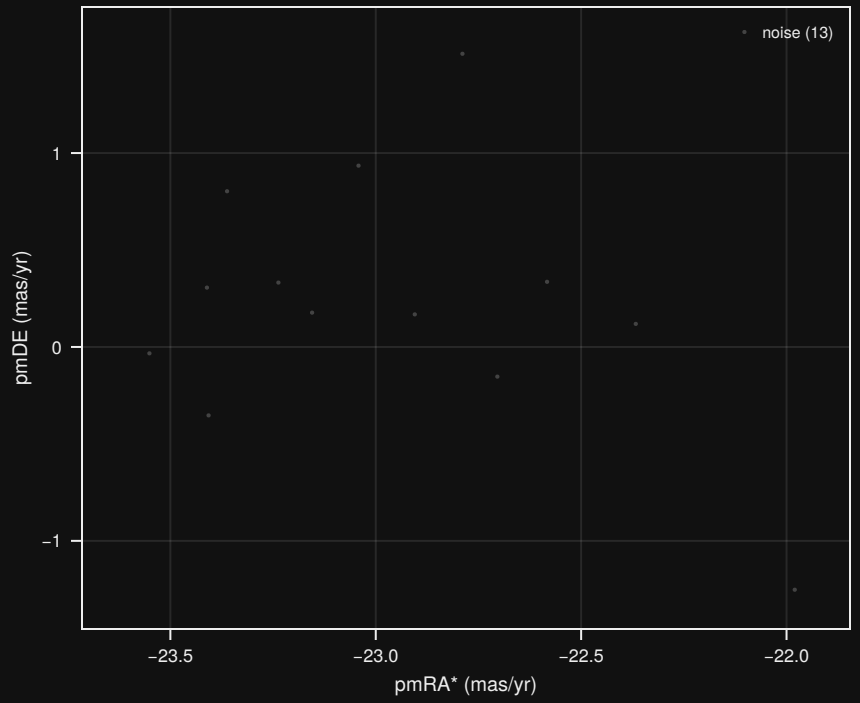


Parallax quality (n = 7)

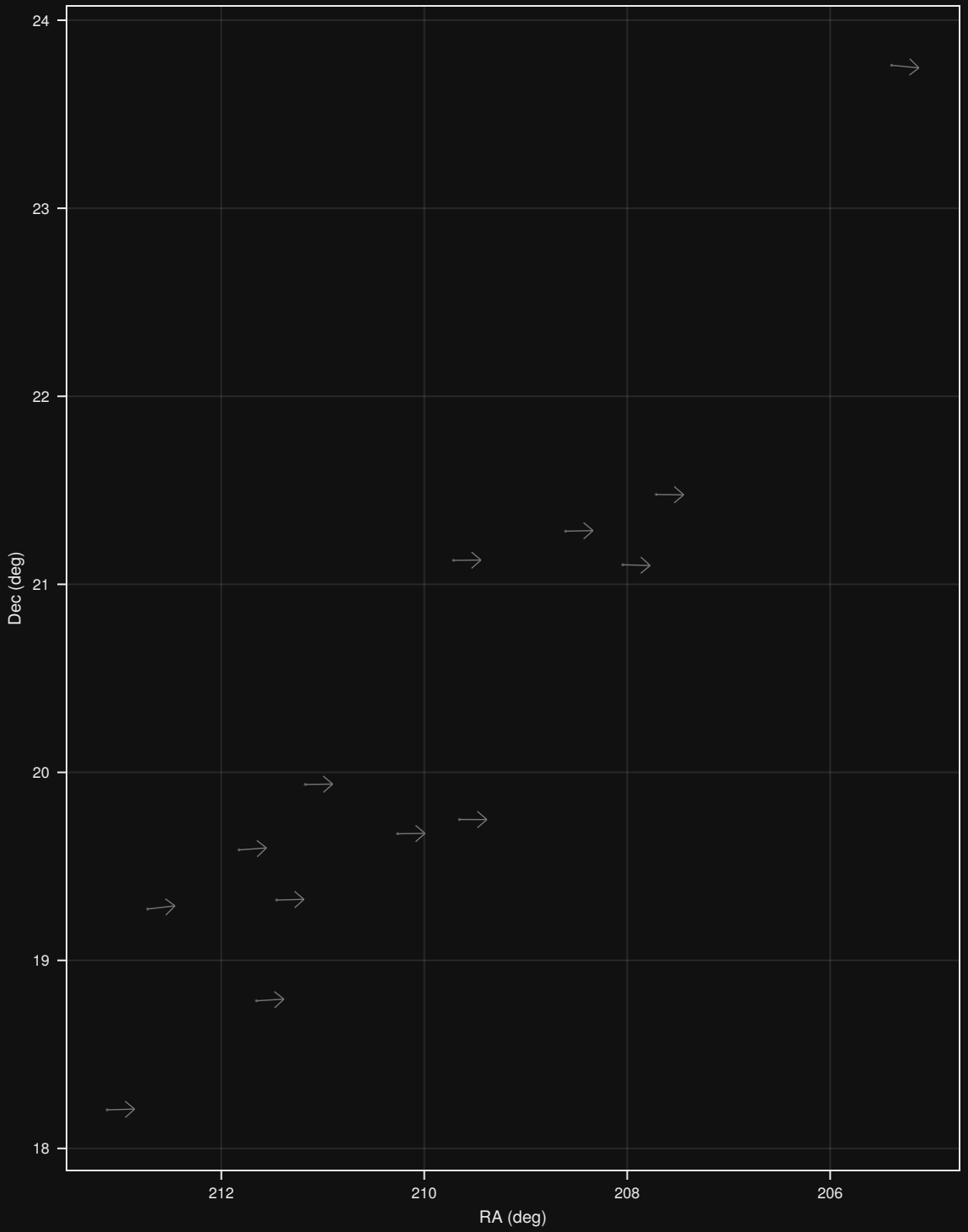


Ophiuchus

PM clustering · HDBSCAN · 0 cluster(s) · noise 100%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	13	-23.04	0.18	0.50	-1.25	0